

Lecture 17: Double Integrals over General Regions

Background & Motivation

Last time we integrated over rectangles. Most regions of interest—disks, triangles, regions between curves—aren't rectangular. Today we handle general regions using Type I and Type II setups, and introduce polar coordinates for circular symmetry. The Gaussian integral, which cannot be evaluated by single-variable methods, yields beautifully to the polar approach.

1. Warm-Up: Double Integral over a Rectangle

Evaluate $\int_0^2 \int_1^3 (3xy^2) dy dx$.

1. Inner integral (y): $\int_1^3 3xy^2 dy =$ _____
2. Outer integral (x): $\int_0^2 26x dx =$ _____
3. Result: _____

2. Type I Regions (Vertical Slicing)

Definition 1: Type I Region

$$R = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

$$\iint_R f(x, y) dA =$$

- Integrate y first (inner), then x (outer)
- Think: vertical lines sweep left to right across the region

3. Type II Regions (Horizontal Slicing)

Definition 2: Type II Region

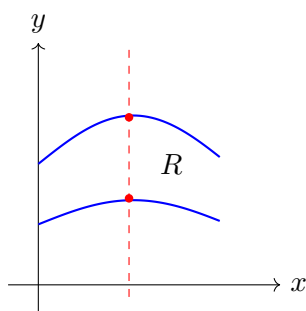
$$R = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

$$\iint_R f(x, y) dA = \underline{\hspace{2cm}}$$

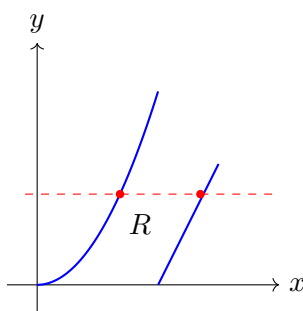
- Integrate x first (inner), then y (outer)
- Think: horizontal lines sweep bottom to top across the region

How to Classify a Region

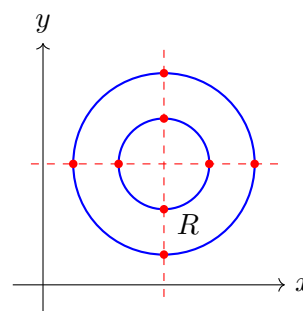
- Sketch the region and find all intersection points
- Type I – can every vertical line ($x = c$) enter through one curve and exit through one curve?
- Type II – can every horizontal line ($y = c$) enter through one curve and exit through one curve?
- Some regions work as both; pick whichever is simpler
- If neither works cleanly, **split** the region



Type I Region



Type II Region



Neither Type

A Type I region is bounded above and below by functions of x — every vertical line (red) hits the boundary exactly twice. Type II is analogous with horizontal lines. An annulus is *neither*: both horizontal and vertical slice lines can cross the boundary four times, so it must be split.

4. Examples

Example 1: Type I: Region Between Two Parabolas

Evaluate $\iint_R (x + 2y) dA$ where R is bounded by $y = x^2$ and $y = 2 - x^2$.

Example 2: Polar: Integral over a Disk

Evaluate $\iint_D (x^2 + y^2) dA$ where D is the disk $x^2 + y^2 \leq 4$.

5. Polar Coordinates for Double Integrals

Double Integrals in Polar

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta$$

- The extra factor of r comes from the area element: $dA = r dr d\theta$
- Use polar when the region or integrand involves $x^2 + y^2$, circles, or sectors

Example 3: The Gaussian Integral (I^2 Trick)

Evaluate $I = \int_{-\infty}^{\infty} e^{-x^2} dx$.

6. Complex Regions (Splitting)

When a region is neither purely Type I nor Type II – **split** it:

$$\iint_R f dA = \iint_{R_1} f dA + \iint_{R_2} f dA$$

You can even use Cartesian for one piece and polar for another.

Example 4: Split Region: Cartesian + Polar

Evaluate $\iint_R (1 + x^2 + y^2) dA$ where R is the region inside the square $[-1, 1] \times [-1, 1]$ but outside the unit disk $x^2 + y^2 < 1$, combined with the upper half-disk $x^2 + y^2 \leq 1, y \geq 0$.

Practice Problems

Problem 1. Evaluate $\int_0^1 \int_0^x xy dy dx$.

Problem 2. Evaluate $\iint_R xy \, dA$ where R is bounded by $y = 0$, $y = \sqrt{x}$, and $x = 4$. (Type I)

Problem 3. Evaluate $\iint_D e^{-(x^2+y^2)} \, dA$ where D is the annulus $1 \leq x^2 + y^2 \leq 4$. (Use polar.)

Problem 4. Reverse the order of integration and evaluate: $\int_0^1 \int_0^{\sqrt{y}} f(x, y) \, dx \, dy$ where $f(x, y) = e^{y/x}$.

Problem 5. Evaluate $\iint_R \sqrt{x^2 + y^2} \, dA$ where R is the region in the first quadrant between $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$.